Section A

1 **(a)** **(i)** $(50 \text{ to } 200) \times 10^{-3}$ kg **A1**

(ii) diameter of tennis ball is about 6 – 7 cm

volume = $\frac{4}{3}\pi r^3$ from 110 to 180 cm^3 **A1**

(b) units of Q: As **C1**

$$C = \frac{Q^2}{2E}$$

units of C = $\frac{\text{A}^2\text{s}^2}{\text{kgm}^2\text{s}^{-2}}$ = $\text{kg}^{-1}\ \text{m}^{-2}\ \text{A}^2\ \text{s}^4$ **A1**

2 **(a)** **(i)** **1** equal **B1**